

Socioeconomic and Human Mobility Datasets of the Rio de Janeiro Metropolitan Area

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Abstract

This article describes two datasets, one on human mobility in 2014 and another on socioeconomic characteristics of the Rio de Janeiro Metropolitan Area (RJMA) in 2010. One of the groups (mobility) was generated from raw data from mobile phone records, and the other (census) is the result of aggregations of socioeconomic data referring to the same coverage area of the mobility data.

Secondary data, covering the same geographic partitioning, compose the two datasets, binded by 55 regions in the RJMA. The mobility bases are arranged over 363 days, covering 867,975 mobility observations on 18 variables. The mobility dataset considers four mobility shifts per day, presenting 3,321,707 observations. The census dataset includes 54 observations, one for each region, on 55 variables. The mobility data was originated from a mobile phone carrier. Seeking to represent the total population, we present two expansion factors. The reuse of the datasets can be potentially helpful for observing human mobility due to event analysis, simulating the spread of contagious diseases, and comparing post-pandemic population mobility behavior, for example.

Keywords: Public data, big data, cities, epidemiology

1. Specification table

Subject area	Data Science.
Specific subject	Human mobility inference from mobile phone records in the RJMA.

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Data types	dta (Stata), Tab-Delimited and RData format.
How the data was acquired	On the mobility dataset, each phone call made from a cellular device records a geospatial trace for billing purposes, called CDR - Call Detail Record. The data described here were produced by aggregation algorithms of CDR data from a mobile phone carrier. On the census dataset, the microdata were extracted from the IBGE public portal and aggregated considering the same geographic partitioning of the mobility data.
Data format	Analyzed.
Description of the data collection	We obtained the CDR raw data from a mobile phone carrier in the context of a research project in urban mobility enrolled in FAPERJ (Fundação de Amparo à Pesquisa do Estado do Rio de Janeiro). We received the data monthly, along with an up-to-date list of mobile phone system service towers with geographic position denoted in latitude/longitude (WGS 1984). At each data load, it was necessary to assign the correct position (Lat/Long) to each CDR record, as some antennas turn into maintenance status or even change their position to meet sporadic demands. Each time new data package was received, it was necessary to eliminate duplicate entries generated by the phone towers. Before cleaning the data, the same CDR was appeared seven times in the database.

Data source location	The raw CDR data on the mobility dataset are not available, due to users' privacy compliance. The phone carrier that provided the data cannot be revealed due to the confidentiality agreement signed by the authors. Data were produced and aggregated in two ways: 1) by day, origin, destination and number of trips; 2) by day, origin, destination, number of visits and shifts, which is described by four time intervals: MORNING (between 4 am and 10 am), WORK (between 10 am and 4 pm), AFTERNOON (between 4 pm and 10 pm) and NIGHT (between 10 pm and 4 am the next morning). Chapter 4, p. 63 in [1] describes the methodology used to produce the data. Census data were extracted from the IBGE portal ¹. • <i>Institute: COPPE/UFRJ, FGV.</i> • <i>Region: Metropolitan Region of Rio de Janeiro (phone code 21).</i> • <i>Country: Brazil</i> • <i>Latitude and longitude: area between (-22.127,-44.230) and (-23.064,-42.444) under WGS 1984 standard.</i>
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¹https://ftp.ibge.gov.br/Censos/Censo_Demografico_2010/Resultados_Gerais_da_Amostra/Microdados/, visto em abril de 2022.

Data accessibility	Aggregated data were already published in the Dataverse repository. • FGV Dataverse, https://dataverse.fgv.br/dataverse/RJMA <i>RJMA ShapeFiles</i>, HANDLE https://hdl.handle.net/10438.3/FK2/N78E9A. <i>RJMA Urban Transport Master Plan interview aggregated data</i>, HANDLE https://hdl.handle.net/10438.3/FK2/TYPWCM. <i>Mobility detected from one phone call</i>, HANDLE https://hdl.handle.net/10438.3/FK2/6AA6UL. <i>Mobility detected from two phone calls</i>, HANDLE https://hdl.handle.net/10438.3/FK2/S5ECCO. <i>Holidays</i>, HANDLE https://hdl.handle.net/10438.3/FK2/NXV1VO. <i>The 2010 Brazil census</i>, HANDLE https://hdl.handle.net/10438.3/FK2/ALDYFQ. <i>SQL scripts used to infer human mobility given by call detail records</i>, HANDLE https://hdl.handle.net/10438.3/FK2/MYC76H. CDR raw data are not available under any circumstances. This measure aims to respect the confidentiality agreement signed by the authors. Instructions for downloading aggregated data: Just download the datasets in Dataverse platform.
Related research article	Chaves, J. C., A.H.B. da Silva, M., Evsukoff, A. G. (2021). Gravity Model Parameter Variation during a Long-Term Study using Mobile Phone Data in the Rio de Janeiro Metropolitan Area. <i>Case Studies on Transport Policy</i>. https://doi.org/10.1016/J.CSTP.2021.12.017

2. Value of the data

- The inference of daily trips between the origins and destinations (O-D) of the RJMA over a year shows a detailed picture of the profile of population mobility in the region. Mobility is presented in two formats: 1) inference of daily trips for each O-D pair from two phone calls and 2) detection of population presence by region and by presumed place of residence over four shifts per day.
- Researchers in the field of transport and in the field of epidemiological models can use the data to train models that take into account the behavior of population mobility under various contexts.
- Researchers in the field of transportation can train parameters of gravitational models or analyze mobility due to significant events, such as New Year's Eve in Copacabana.
- Transportation models represent one of the biggest challenges when evaluating epidemiological models, especially in Brazil, where the scarcity of public mobility data with a defined scope and methodology makes the task even more challenging. Moreover, researchers in the field of epidemics can discover insights about movement patterns between regions under the daily course of several days and shifts, as well as changes in these patterns due to weekdays, weekends, holidays and long holidays.

3. Objective

This data article presents algorithmically computed human mobility data from cell phone call records (CDR) in the Rio de Janeiro Metropolitan Area (RJMA). This dataset represents a long-term study of daily mobility data in four shifts, totaling 3,321,707 travel records (origin and destination) between the 55 geographic units of the RJMA in 2014. The study uses spatial modeling to approximate the geographic units, which allows for easy integration with data from other sources. The dataset also includes socioeconomic data from the 2010 Census and data from 2013 RJMA Transportation Plan Survey data from 2013, which were used to validate the results obtained from the CDR. This descriptor complements the results presented in the [2] article so that they can be replicated and also serves as support for other studies, such as those in the field of epidemiology.

4. Data description

This article describes two secondary datasets, with population mobility data as the main set and census data as an auxiliary layer. In order to facilitate geospatial analysis, maps are available in Shapefile format for each set. Furthermore, in order to facilitate the use of statistical tools, data is

shared in ~~two~~three^{jcc} formats: 1) dta (Stata), 2) tsv (tab separated file). The data coverage area extends mainly to the RJMA and some adjacent municipalities. The mobile operator refers to it as Area 21 in Brazil, as seen in the Figure 1.

The default geographic unit is the municipality, except for the municipality of Rio de Janeiro, which has been partitioned into sub-districts to balance the study area's population density. Therefore the datasets include more than one city.^{jcc}

[Figure 1 about here.]

Each region can be studied based on its individual objectives, in accordance with the characteristics of interest for the study. The demographic census will consider socio-economic attributes of the population, so it is sensible to segment populations with homogeneous socio-economic qualities. Furthermore, the transport area is concerned with the mobility of the population, thus it is logical to define boundaries according to traffic zones that take into account the road network (Problems, Study Areas, Models, And Data, in [5]).^{jcc}
novo^{jcc}

In order to illustrate the practical implications of linking databases with one another, we selected a set of census variables that are well-known across a variety of domains and which are used to calculate the Human Development Index ². These variables may be combined by experts to produce indicators related to education, life expectancy, and income indicators, as well as gender, race, and mobility analysis.

The definition of geographic partitioning was guided to overcome ~~three~~two^{jcc} obstacles: 1) to allow the link with the socio-economic data of the Brazilian census, 2) to represent the coverage areas of the phone towers, and 3) to represent valid traffic zones within the study area^{jcc}.

In general, at least three antennas are oriented at 120° on each tower so that Voronoi polygons can approximate coverage per tower. Figure 2 in [2] shows a cropped view of the study area showing the spatial distribution of the antennas, their respective Voronoi polygons, and aggregated regions.

The positioning of the towers follows marketing and technical objectives and their distribution is quite irregular. The spatial distribution of antennas is denser in areas of higher income and economic activity. In the North and West areas, with lower population density, the distribution of antennas is less dense, as well as in the other municipalities in the Metropolitan Region, except for the center of Niterói. The low density of phone towers observed in some areas in the middle of the urban area is

²HDI - <https://hdr.undp.org/data-center/human-development-index#/indicies/hdi>.

justified by the presence of mountains. The geography of the city of Rio de Janeiro has a significant impact on the city’s mobility routes.

Figure 3 in [2] shows the arrangement of Voronoi polygons over some sub-districts of the city of Rio de Janeiro. In regions with a higher density of antennas, it is easier to respect the boundaries between sub-districts. However, the approximation in low-density regions tends to be less accurate, even coherent with the uncertainty of coverage in such vast regions.

The data described in this article were analyzed from different perspectives. A methodological study regarding the field of transportation was presented in [3]. A study on parameters estimation for a standard model in the field of transportation, the Gravitational Model, can be found in [2]. Moreover, an investigation of the relationship between geospatial inequalities and economic aspects considering the entropy of attractiveness in different regions was presented in [4].

4.1. Populational mobility data in the RJMA

4.1.1. Dataset 1 - Mobility detected from two phone calls

Filename: “RJMA_ordinary_mobility_given_by_two_calls”.

This file contains the typical human mobility along 2014 obtained from two consecutive calls. Between origins and destinations, there are three different travel counts: Travel without expansion factors, travel with fixed expansion, and travel with adaptive expansion.

Table 2: Aggregated mobility dataset, defined from two mobile phone calls.

Variable	Type	Description	Domain
day	Date (dd/m- m/yyyy)	Date when the phone call occurred, and consequently, the trips were produced.	363 days in 2014.
origin	String(20)	Geographical unit of origin of trips.	54 geographic partitions.
destination	String(20)	Geographical unit of destination of trips.	54 geographic partitions.
travell_counting_no_expansion	Integer	Number of trips detected directly from the mobile phone carrier.	Natural numbers.

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Table 2 – continued from previous page

Variable	Type	Description	Domain
travell_counting_ fixed_expansion	Integer	Number of trips expanded in fixed way – <i>k-fixed</i> , according to Section 4.3.1, p. 75 in [1]	Natural numbers.
travell_counting_ adaptive_expans	Integer	Number of trips expanded in adaptive way – <i>k-adaptive</i> , according to Section 4.3.1, p. 75 in [1]	Natural numbers.

4.1.2. Dataset 2 - Mobility detected from one phone call

This file includes data on human mobility from 2014 that was gathered from a single call. Origins are assumed domiciles, while destinations are locations where a cell phone user makes a single call and establishes their presence. For the purpose of observing how the population behaves in relation to the seasons, holidays, and shifts, various categorical attributes are made available.

Filename: “RJMA_ordinary_visiting_given_by_one_call_over_shifts”.

Data dictionary: See appendix in Section 6.1.

The dataset in Stata format is shared with the `label` feature, replacing several categorical text variables with `byte` variables, considerably reducing the memory consumption for manipulating the application.

4.1.3. Dataset 3 - RJMA Urban Transport Master Plan (PDTU - Plano Diretor de Transporte Urbano)

This file contains commuting aggregated data that was gathered through official government interviews with a sample of the public as part of the initiative to develop an urban transportation plan (PDTU).

Filename: “Commuting_given_by_Transportation_Interviews_2013”.

RJMA’s PDTU is the official reference for studies in the field of transportation, made by the Secretary of Transport of the State of Rio de Janeiro. The PDTU microdata are available at the level of traffic zones, making aggregation possible at the sub-district level and consequently at the municipality level.

Table 3: Aggregated dataset from the PDTU in the RJMA in 2013.

Variable	Type	Description	Domain
origindestination	String(50)	Origin and destination geographic units in “Origin-Destination” format.	All pairs of geographic unities in the study area combined.
commuting_by_interviews_2013	Integer	Number of trips under “Origin-Destination” obtained from surveys in 2013.	Natural numbers.

4.1.4. Dataset 4 - ShapeFile RJMA_21_area-traffic-zones-partitioning

This ShapeFile depicts the study area’s partitioning scheme, revealing sub-districts, municipalities, and macro traffic zones .In addition to the geographic partition name, the region code from the Brazilian demographic census is displayed. This measure aims to make it easier to connect to other databases.

Table 4: Shapefile of area 21 containing traffic macro-zones.

Variable	Type	Description	Domain
grao	String(11)	Code of the sub-district or municipality in the IBGE standard.	55 geographic partitions.
pop_usu	Integer	Population of phone users detected in the region (grao).	Natural numbers.
pop_censo	Integer	Population declared by the 2010 census in the region (grao).	Natural numbers.
nm_grao	String(50)	Name of sub-district or municipality.	55 geographic partitions.
macrozona	String(50)	Code of the subdistrict or municipality in the IBGE standard.	12 geographic partitions.

4.1.5. Dataset 5 - ShapeFile RJMA_21_area_union_set_towers

Denotes the polygons resulting from the aggregations of the coverage areas of the phone towers inferred from Voronoi polygons.

Table 5: Shapefile of area code 21 denoting geographic partitions as aggregations of Voronoi polygons from phone towers.

Variable	Type	Description	Domain
nm_grao	String(50)	Name of the sub-district or municipality.	55 geographic partitions.
census_id	String(11)	Code of the sub-district or municipality in the IBGE standard.	55 geographic partitions.

4.1.6. Script - human_mobility.sql.tex

This script shares all of the SQL commands that were used to generate the data. The file has several instructions. By means of a guided presentation of the instructions, the file is presented with comments in Latex standard.

4.2. Census data in the RJMA

The linked and arranged census files of the *ShapeFile* type provide geographically located socioeconomic information about the study area, which can be used to visualize relationships between variables. For example: 1) Figure 2 shows population, education and socioeconomic indicators.

[Figure 2 about here.]

4.2.1. Dataset 6 - Socioeconomic data of the study region

Filename: “RJMA_census_2010”. Data dictionary: See appendix in Section 6.2.

This dataset exemplifies how one can link data from various sources for exploratory analysis of the region using the “census.code”. Several socio-economic aspects of the study area, including population and household, are depicted.

5. Experimental design, materials and methods

The study made use of a CDR dataset that was gathered from 1078 antennas in the Brazilian area code 21. The information, which included 2.1 billion call records generated by 2.9 million customers, was compiled between December 31, 2013, and January 1, 2015. The dataset for the study did not include any information about received calls, text messaging (SMS), or mobile internet usage; only information about outbound voice calls was available. Based on an analysis of the distribution of the number of calls, the data were cleansed to remove users who exhibited unusual behaviour.

Data were produced by executing sequences of standard SQL instructions on raw CDR data and demographic census data. All SQL scripts are being shared along with the data (see Section 4.1.6). The census data provided the structure without which it would not be possible to define what characterizes the geographic regions and, consequently, the possible origins/destinations. The raw CDR data reflect a standard coordinate system compatible with most popular systems, such as the WGS 84 used by Google Maps. The definition of a standard geographic partitioning was a prerequisite that influenced and made possible the production of all data since it deals with the concept of what is a geographically delimited area. Based on this foundation, a more usual sequence of steps for research in the area of mobility was followed, inferring the users' place of residence and applying expansion factors, to finally infer the population mobility itself.

Ethical declarations

All private data were encrypted and there was no Ethical Committee evaluation. This measure aims to preserve the privacy of users under the protection of the General Data Protection Law (LGPD - *Lei Geral de Proteção de Dados*) and comply with the confidentiality agreement signed by the authors. The derived data, described and shared here, do not identify any natural person.

CRedit author statement

Julio C. Chaves: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Validation, Resources, Visualization. **Moacyr A.H.B. da Silva:** Conceptualization, Methodology, Funding acquisition, Validation, Formal analysis, Investigation, Writing – review & editing, Project administration. **Alexandre G. Evsukoff:** Conceptualization, Methodology, Data curation, Funding acquisition, Writing – original draft, Validation, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration. **Ricardo de Souza Alencar:** Validation, Formal analysis, Writing-review & editing. **Vinícius da Fonseca Vieira:** Investigation, Validation, Writing-review & editing.

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Declaration of Interest

- ✓ The authors declare that they have no known competing financial interests or personal relationships that may have influenced the work reported in this article.

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6. Appendix

6.1. Filename: “*RJMA_ordinary-visiting-given-by-one-call-over-shifts*”.

The dataset in Stata format is shared with the `label` feature, replacing several categorical text variables with `byte` variables, considerably reducing the memory consumption for data manipulation.

Table 6: Aggregate mobility dataset in RJMA in 2014, produced from a mobile phone call.

Variable	Type	Description	Domain
dayid	Date (dd/mm/yyyy)	Day when phone calls occur and consequently the trips were detected.	363 days in 2014.
day_descrid	String(33)	Description of the day, detailing holidays, Brazil 2014 world cup games or simply normal working days.	63 description, eg.: {"ALE x ARG (FINAL)", "Carnival", "Reveillon",...}
day_typeid	String(11)	Brief description of the day, highlighting only if it is a working day, a holiday or a weekend.	{"holiday", "non-workday", "work-day"}
seasonid	String(9)	The season.	{"AUTUMN", "SPRING", "SUMMER", "WINTER"}
shiftid	String(9)	The day shift. There are 4 shifts of 6 hours each: (MORNING) 4 am - 10 am, (WORK) 10 am - 4 pm, (AFTERNOON) 4 pm - 10 pm, (NIGHT) 10 pm - 4 am. For reference, see [2].	{"AFTERNOON", "MORNING", "NIGHT", "WORK"}
domicileid	String(20)	Geographical unit of residence of phone users who have made at least one call in <i>destinationid</i> .	54 geographic partitions.
destinationid	String(20)	Geographic unit where users residing in <i>domicileid</i> made at least one call, characterizing a visit.	54 geographic partitions.

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Table 6 – continued from previous page

Variable	Type	Description	Domain
census_ popula- tion_ from_ l	Integer	Number of people residing in the geographic partition of residence <i>domicileid</i> according to the 2010 IBGE census.	Natural numbers.
cell.phone_ users_from_l	Integer	Number of phone users of a single phone carrier residing in the geographic partition of residence <i>domicileid</i> .	Natural numbers.
total_detected_ user_from_l	Integer	Number of phone users of a single phone carrier residing in the geographic partition <i>domicileid</i> who have made at least one phone call in the destination <i>destinationid</i> .	
visiting_population at_destination	Integer	Population residing in <i>domicileid</i> inferred at destination <i>destinationid</i> from the expansion of a phone call at the destination, according to Section 4.5, p. 88 of [1].	Natural numbers.
estimated_distance travelled_met	Integer	Traveled distance estimated (in meters) between the centroids of the source and destination geographic partitions by access roads ³ .	Natural numbers.

6.2. Filename: “RJMA_census_2010”.

³Values obtained with Google Maps Distance Matrix API.

Table 7: RJMA aggregated socioeconomic dataset in 2010. Source: IBGE 2010 census.

Variable	Type	Description	Domain
census_code	String(11)	Region identifier. The code with 7 positions refers to the municipality. The code with 11 positions refers to the sub-district	Cities or sub-districts.
city_or_subdistrict	String(20)	Geographic partition name.	Cities or sub-districts.
adult_can_read_and_write	Number of adults who can read and write.	Natural numbers.	
young_can_read_and_write	Integer	Number of young people who can read and write.	Natural numbers.
men_no_education	Integer	Number of men without complete primary education.	Natural numbers.
women_no_education	Integer	Number of women without complete primary education.	Natural numbers.
men_primary	Integer	Number of men with complete primary education, or with incomplete secondary education.	Natural numbers.
women_primary	Integer	Number of women with complete primary education, or with incomplete secondary education.	Natural numbers.
men_high_school	Integer	Number of men who have completed high school, or have incomplete graduation.	Natural numbers.
women_high_school	Integer	Number of women who have completed high school, or have incomplete graduation.	Natural numbers.

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Table 7 – continued from previous page

Variable	Type	Description	Domain
men_undergraduated	Integer	Number of men who have completed graduation.	Natural numbers.
women_undergraduated	Integer	Number of women who have completed the graduation.	Natural numbers.
men_unknow_education	Integer	Number of men with unknown educational level.	Natural numbers.
women_unknow_education	Integer	Number of men with unknown educational level.	Natural numbers.
men_deceased	Integer	Number of males over 5 years old who has died.	Natural numbers.
women_deceased	Integer	Number of females over 5 years old who has died.	Natural numbers.
child_deceased	Integer	Number of children up to 5 years old who have died.	Natural numbers.
avg_dormitories	Real	Average bedrooms per housing unit	Real numbers.
water_outside_house	Integer	Number of housing units with water on the property, but outside the house.	Natural numbers.
no_water	Integer	Number of housing units without water.	Natural numbers.
water_in_the_house	Integer	Number of housing units with water inside the house.	Natural numbers.
sewage_ditch	Integer	Number of housing units whose sanitary drain is connected to a ditch.	Natural numbers.
sewage_sceptic_tank	Integer	Number of housing units whose sanitary drain is connected to a septic tank.	Natural numbers.
sewage_rud_sceptic_tank	Integer	Number of housing units whose sanitary drain is connected to a rudimentary septic tank.	Natural numbers.

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Table 7 – continued from previous page

Variable	Type	Description	Domain
sewage_river_etc	Integer	Number of housing units whose sanitary drain is connected to a river, lake or ocean.	Natural numbers.
sewage_network	Integer	Number of households with a sanitary drain connected to the public sewage system.	Natural numbers.
population	Integer	Total population of the region.	Natural numbers.
pct_male	Real	Percentage of the population of men.	Real numbers.
pct_female	Real	Percentage of the population of women.	Real numbers.
xwomen_for100men _above15	Integer	How many women count for every 100 men (both over 15)?	Natural numbers.
pct_pop_above_15	Integer	Percentage of population over 15 years old.	Natural numbers.
xmen_for100women _above60	Integer	How many men count for every 100 women (both over 60)?	Natural numbers.
pct_pop_above_60	Integer	Percentage of population over 60 years old.	Natural numbers.
pct_white	Real	Percentage of residents who declared their color or race white.	Real numbers.
pct_black	Real	Percentage of residents who declared their color or race as black.	Real numbers.
pct_asian	Real	Percentage of residents who declared their color or race as yellow (Asians).	Real numbers.
pct_brownbrazil	Real	Percentage of residents who declared their color or race as pardos (mulatto).	Real numbers.
pct_indigenous	Real	Percentage of residents who declared their color or race as indigenous.	Real numbers.
man_income	Real	Total monthly income per man (average).	Real numbers.

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Table 7 – continued from previous page

Variable	Type	Description	Domain
woman_income	Real	Total monthly income per woman (average)	Actual numbers.
housing_percapita	Real	Per capita income per housing unit. and real numbers.	
men_employed	Integer	Number of men employed.	Natural Numbers.
women_employed	Integer	Number of women employed.	Natural numbers.
men_employee	Integer	Number of male employers.	Natural numbers.
women_employee	Integer	Number of women employed.	Natural numbers.
pct_works_in_home	Real	Percentage of residents aged 10 years or older over who work from their own home.	Real numbers.
pct_works_in_municipality	Real	Percentage of residents aged 10 years or older who work in the municipality, but not at home.	Real numbers.
pct_works_in_another_municipality	Real	Percentage of residents aged 10 years or older who work in another municipality.	Real numbers.
pct_works_in_foreign_country	Real	Percentage of residents aged 10 or older working in a foreign country.	Real numbers.
pct_commute_hw_5min	Real	Percentage of residents aged 10 or older who take up to 5 minutes to commute to work.	Real numbers.
pct_commute_hw_b6min_and_half_h	Real	Percentage of residents aged 10 or older who take 6 minutes to half an hour to go to work.	Real numbers.
pct_commute_hw_bhalf_h_and_1h	Real	Percentage of residents aged 10 or older who take half an hour to an hour to commute to work.	Real numbers.

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Table 7 – continued from previous page

Variable	Type	Description	Domain
pct_commute_hw_bReal _1h_and_2h		Percentage of residents aged 10 or older who take more than one to two hours to commute to work.	Real numbers.
pct_commute_hw _more_than_2h	Real	Percentage of residents aged 10 or older who take more than 2 hours to com- mute to work.	Real numbers.
pct_return_from _work_everyday	Real	Percentage of residents aged 10 or older who return home from work each day.	Real numbers.

List of changes

Replaced (jcc): two	6
Added (jcc): Therefore the datasets include more than one city.	6
Added (jcc): Each region can be studied based on its individual obSS [...]	6
Added (jcc): novo	6
Replaced (jcc): three	6
Added (jcc): and 3) to represent valid traffic zones within the study area	6

List of Figures

1	Data coverage area comprising the RJMA. Each polygon is a geographic partition, from 1 to 54.	21
2	An example of the visualization of socioeconomic data for the region: (a) population density; (b) average income in (R\$) real; (c) percentage of residents who declared themselves to have completed secondary education, or higher;(d) percentage of residents who declared themselves to have completed elementary school, or less.	22

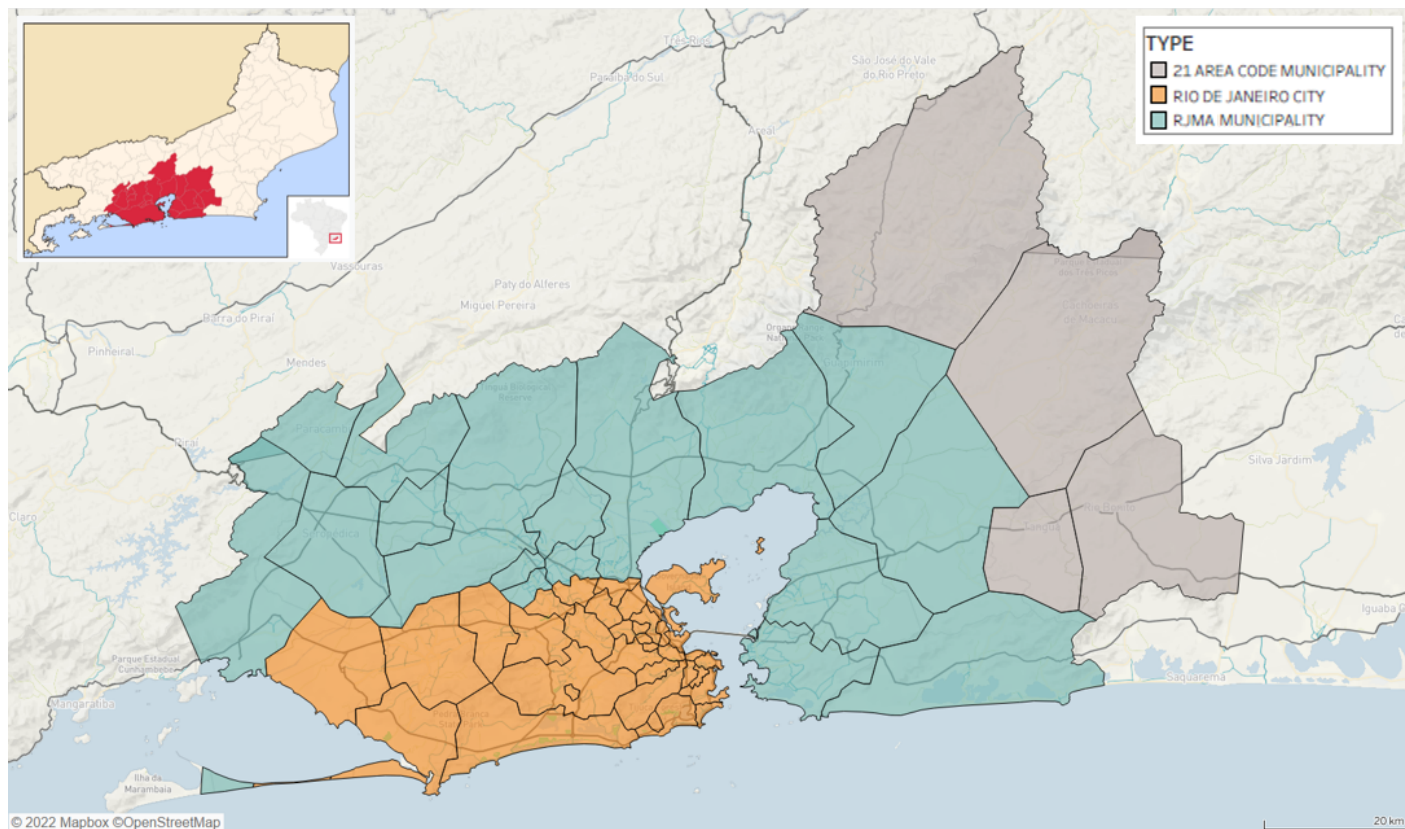
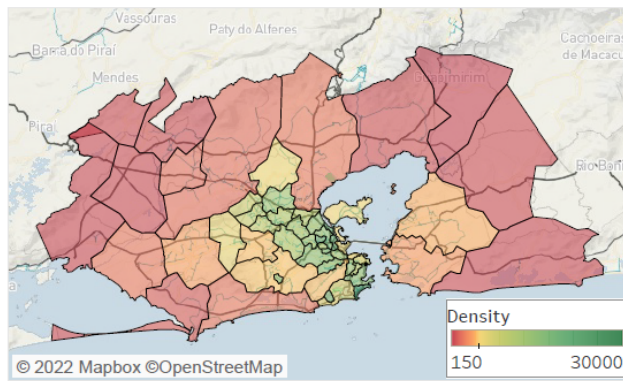
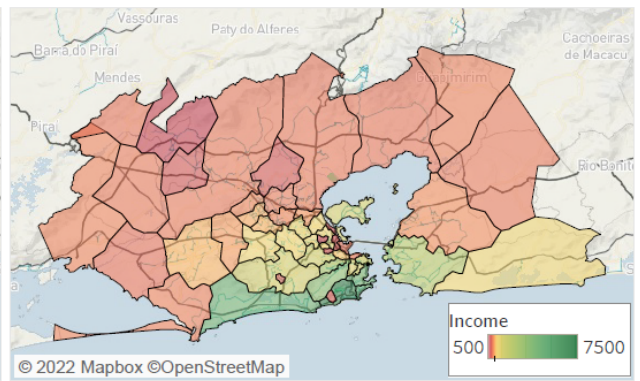


Figure 1: Data coverage area comprising the RJMA. Each polygon is a geographic partition, from 1 to 54.

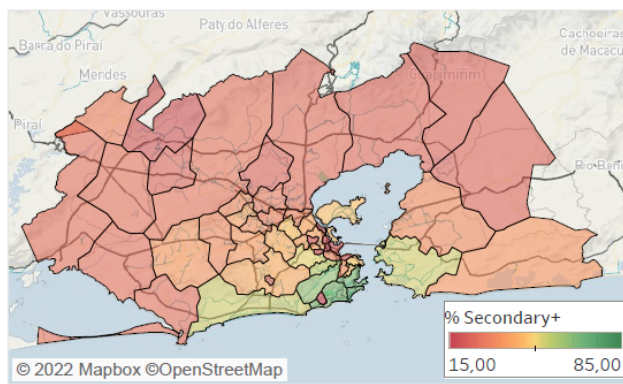
(a) DENSITY



(b) INCOME



(c) % SECONDARY +



(d) % ELEMENTARY -

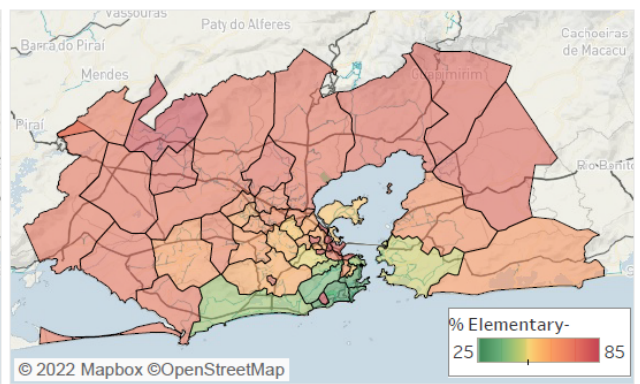


Figure 2: An example of the visualization of socioeconomic data for the region: (a) population density; (b) average income in (R\$) real; (c) percentage of residents who declared themselves to have completed secondary education, or higher;(d) percentage of residents who declared themselves to have completed elementary school, or less.